

SVD

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September 7, 2025

Task 1 (2p)

For this exercise I selected Indian pines as my dataset. The dataset was loaded in MATLAB and the constructed RGB image and false color image at 500 nm are shown in Figure 1. A spectrum of pixel at [3,4] is shown in Figure 2.



Figure 1: RGB image and the false color at 500 nm band

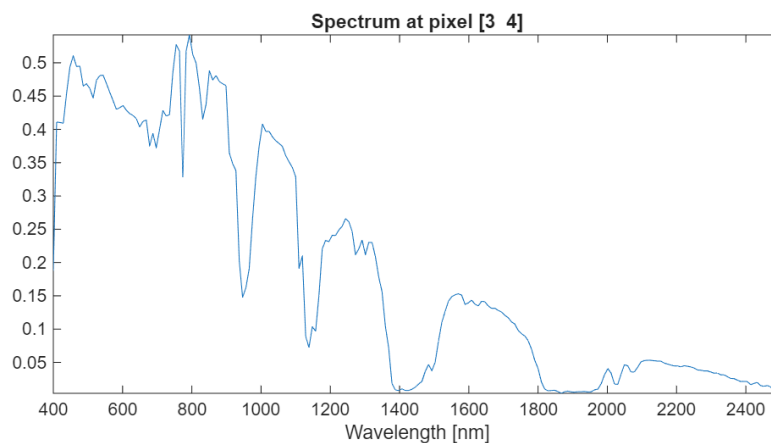


Figure 2: Spectrum at pixel [3,4]

The dataset dimensions are $145 \times 145 \times 220$ so the total number of cells required for storing the data is $145 \cdot 145 \cdot 220 = 4625500$. The dataset was reshaped to 2D matrix for the following task.

Task 2 (2p)

Economy SVD was used with the reshaped dataset matrix. The dataset was normalized before SVD in order to have zero mean and unit variance. The explained variance of the singular vectors was calculated as

$$\text{Var}(\lambda_i) = \frac{\lambda_i^2}{\sum \lambda_j^2} \quad (1)$$

where λ is a vector containing all singular values. Cumulative explained variance of the singular vectors are shown in Figure 3.

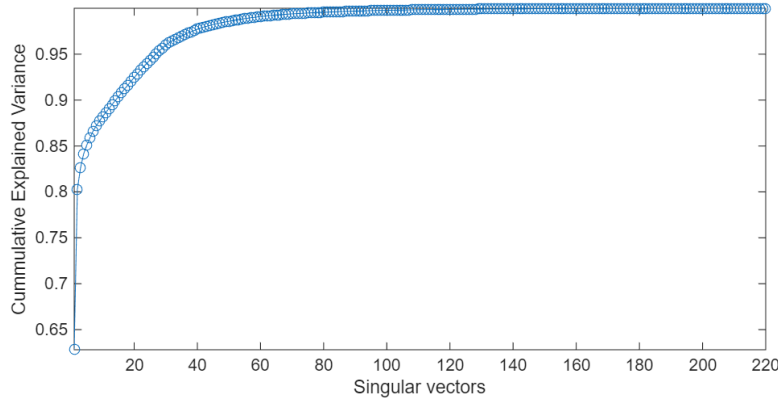


Figure 3: Cumulative explained variance of the singular vectors

If we use 95% as the threshold for determining the reconstruction rank, we get $R = 27$ as the rank based on the cumulative explained variance. The matrices were truncated using R for the following task.

Task 3 (3p)

Dimension of the truncated US is $21025 \times 27 =$ and the dimension of the truncated V is 27×27 . Reduced dimensionality after truncation is hence $21025 \cdot 27 + 27 \cdot 27 = \mathbf{568404}$. When comparing this to the original number calculated in Task 1, we get **87.71%** reduction from the original size for this particular value of R .

The RMSE was calculated by comparing the original data matrix to the reconstructed one. With the R used here we get $\text{RMSE} = \mathbf{16.99\%}$, which is significantly lower than the compression rate. False color image of the reconstruction error is constructed by calculating the relative error to the original dataset and summing all the wavelengths bands together as a one 145×145 layer. The reconstruction error image is shown in Figure 4.

In the image we can see a few individual spots (in yellow color) where the error is relatively larger than elsewhere in the image. These spots seem random, and can be caused by some anomaly in the ground level that was not captured in the larger singular values.

Most of the outlines of the original image can also be seen in the error image, which means that some areas in the image are more accurate in the reconstructed image than some other areas. This type of error seems systematical, and can be caused by for example different type of pine



Figure 4: False color image of the reconstruction error

trees in the areas. In the image we can see, that in the center there are large areas with very little error values, but some of the surrounding areas have larger error values.